AI Journals

Julia Kay Gutiza - 170706L

Section 1: AI Tools Declared

NotebookLM (Google) – Used to extract and organise conversion equations from uploaded research papers, cross-reference sources, and identify which papers covered bidirectional conversions and error cases. Also used the audio overview feature to understand the basics before diving into the equations.

Claude (Anthropic, Sonnet 4.6) – Used throughout the project for interpreting the provided material from NotebookLM, structuring the Jupyter notebook, debugging code errors, writing and formatting the written notes document, reviewing the simulator's mathematical consistency, and resolving Git merge conflicts.

Section 2: Notable Prompts

NotebookLM research session on conversion equations.

Prompts: *"What do these papers say breaks or causes errors during colour space conversion?", "Do any of the papers provide the inverse conversion equations?", "Do the papers describe converting between non-RGB spaces like HSV to LAB directly?"*

These three prompts were used in NotebookLM after uploading the research papers to extract specific information that would have taken significantly longer to find by reading each paper in full. The first returned a detailed breakdown of error types: rounding, integer truncation, singularities, and geometric distortions, which helped shape the 'what breaks' section of the written notes and the error demonstration cells in the notebook. The second confirmed that bidirectional conversions were documented in the Muratbekova and Ford & Roberts sources, which justified including inverse conversions as part of the conversion section rather than treating them as out of scope. The third confirmed that no direct HSV to LAB formula exists in any of the sources, which provided a citable basis for the 'RGB as hub' explanation rather than it being an assumption.

Prompts throughout implementation and development of the actual assignment.

Prompt 1: *"i have these notes extracted from a few papers using notebooklm. help me create them in a simple but detailed way such that a person who's hearing this for the first time understands"*

Provided raw extraction from NotebookLM covering all six conversions and asked Claude to help restructure them into clean academic notes. Claude helped reorganise the content into a draft set of notes for the conversions section with analogies, worked examples, a quick reference cheat sheet, and a glossary from the provided research material. It changed the work because it compressed hours of formatting into a single step, though it required checking every equation against the original sources manually afterward and making sure they can be easily understood for a learning student.

Prompt 2: *"RuntimeWarning: invalid value encountered in divide S = np.where(V == 0, 0, delta / V)"*

Pasted the runtime warning from the RGB to HSV cell from the walkthrough notebook. Claude explained that NumPy evaluates both branches of np.where before selecting, meaning the division executes even where V=0, and provided the V_safe fix. This was a debugging moment that taught an important NumPy behaviour that was not obvious from the warning message alone.

Prompt 3: *"the research notes i gave from the sources mention things like floating point, LUT, etc. are those all to be implemented somewhere in the notebook or is it something i should leave to discuss within actual written notes?"*

Asked how to handle the gap between what the literature described and what the notebook actually implemented. Claude explained that integer-based and LUT methods exist for hardware constrained environments and would be meaningless to demonstrate in a Python/NumPy context, and suggested using the environment itself as the justification for choosing the floating-point method. This directly shaped the introductory paragraph of the conversion section.

Prompt 4: *"knowing all this information on conversions from the notes, I need to create 2 multiple choice questions that are not extremely easy to come to the answer but requires a level of understanding"*

Claude generated two multiple choice questions targeting conceptual understanding rather than recall; one on YCbCr offsets and one on colour space selection for real-time detection, each with a worked explanation of why the correct answer is correct and why the distractors are plausible but wrong.

Section 3: 5 Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

My role was focused on the conversion equations, and managing the jupyter notebook walkthrough. The hardest concept was why gamma linearisation is necessary before converting RGB to LAB. RGB values are adjusted for how humans see brightness, not for actual light intensity, while the XYZ/LAB conversion assumes linear light values. Skipping gamma correction means the conversion maths is working with the wrong type of data, so the colours become physically inaccurate rather than just slightly less precise. It mixes physics, human vision, and maths together, so people need to understand several ideas at once instead of just one concept. The error from skipping gamma correction is invisible in the equations themselves as the maths still works, just on the wrong kind of data. It makes the mistake hard to grasp and see.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

When Claude first generated the YCbCr inverse reconstruction code, it used (Y-16) in the inverse equations. This was correct for the integer BT.601 standard, but the forward conversion was still using a normalised version without the +16 offset. The round-trip difference image showed visible errors instead of near-black noise, revealing that the two functions were inconsistent. The fix was to make both the forward and inverse conversions use the same standard.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

Early in organising the notebook, Claude suggested adding channel split visualisations directly inside the conversion sections, such as showing the H, S, and V channels after RGB to HSV conversion. Later, I realised that they better belong in the HSV explanation section rather than the conversion section, since they demonstrate what the colour space represents instead of how the conversion works. Moving them took extra time that could have been avoided by defining the section boundaries more clearly before writing the code, and so I learned to be more specific when prompting in order to get the most relevant information back.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

Instead of manually turning multiple sets of raw equation notes into organised explanations, Claude helped structure the content into clear step-by-step conversion processes with simplified equations, worked examples, and explanations. This made the material much easier to follow and provided a strong foundation that only required minor corrections and refinements. The final structure was also clearer and more logical than the original layout that had been planned manually. We were able to focus on making sure the information was correct rather than fussing over if it was simple enough for a learner to understand.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

The team still cannot determine the exact point at which a floating-point rounding error becomes visually noticeable to a human observer. The difference maps in the notebook show that round-trip conversion errors do occur, but it is unclear whether a specific error value (like a maximum L* difference of 2.3 after a LAB round-trip) would actually be visible in a real image. Delta E is commonly used to measure perceptual colour differences, but the threshold at which a difference becomes noticeable is not fixed. It can vary depending on the colour region, lighting conditions, and surrounding colours in the image. Although the sources suggest that these errors are acceptable for tasks such as skin detection, they do not provide a universal perceptual threshold that applies consistently across all colour conversion methods.

Romey Bajada - 0014306H

Section 1: AI Tools Declared

Claude (Anthropic, Sonnet 4.6) – Used as a personal tutor throughout the project: breaking down difficult concepts, enhancing the structure and vocabulary of the study notes, acting as a proofreader, finding relevant peer-reviewed research papers and brainstorming the simulator's features and functionality.

Claude Code (Anthropic, Opus 4.6) – Used for the initial building of the simulator once its structure and features had been agreed upon, as well as debugging and improvements throughout development.

NotebookLM (Google) – Relevant papers were uploaded and summarised, with key takeaways extracted from each. Particularly useful for identifying real-world applications of colour spaces to include in the learning pack.

Section 2: Notable Prompts

Prompt 1: *"We need to improve the hsv color masking section: Either we add rough estimates of color hues because people are not likely to know off the top of their head that pink is around 300 for example.Also would it make sense to have only one hue slider than apply ±20 for the min and max, making it simpler, or would that reduce functionality?"*

This prompt came at the point where I was assessing the simulator experience from a student's perspective. Claude Code pushed back on the single-slider idea, clearly explaining that simplifying was not worth the loss in functionality. It then improved on my colour label idea by suggesting a labelled colour band alongside the sliders, and independently proposed adding colour presets that the students could select and fine-tune. It asked how I wanted to proceed before generating the implementation code.

Prompt 2: *"What should a simulator for this assignment look like?"*

Asked at a point when a substantial portion of the notes were complete and loaded into context. Claude returned a generic outline that described a basic color converter without highlighting what makes each colour space distinctly interesting. There was no mention of colour masking in HSV, perceptual uniformity in LAB, or chrominance discarding in YCbCr. This response made clear that vague prompts produce vague results, and pushed me toward more targeted prompting for each colour space individually.

Prompt 3: *"Using the hsv notes section as context, what are 10 quiz questions one could ask. Each question should have 4 multiple choice answers. For each question write a justification as to why you picked it and its corresponding answers. Highlight common misconceptions in the topic"*

For the adversarial quiz I was responsible for contributing two HSV questions. I requested ten so I could hand-select the two that best tested genuine understanding rather than memorisation. Claude returned a good spread, some on conversion maths, but mostly on real applications and misconceptions. The selection process itself helped me identify which concepts were most likely to trip students up.

Prompt 4: *"I am compiling a Learning Pack about Colour Models and Colour Spaces (RGB, HSV, YCbCr, LAB. Conversion equations, why each exists, what breaks when you use the wrong one) for students at a university level. For this source, what are the paper's key takeaways and real world applications that should be mentioned in the learning pack?"*

Used with NotebookLM during the early drafting phase, with relevant papers already uploaded. At this stage the focus was purely on content rather than structure. NotebookLM highlighted which concepts were most worth citing. Structure and flow were addressed in a later phase.

Section 3: 5 Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?
Hue instability and why we need saturation thresholds was the hardest concept to communicate. It is difficult because it requires understanding two things: the geometry of what happens at the centre of the HSV cylinder, and why this produces a practical bug in OpenCV pipelines. Connecting the two into a single explanation took several attempts.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?
The first simulator structure idea produced by Claude was technically functional but shallow and generic. It was a colour converter that did not demonstrate the scenarios that justify each colour space's existence. There was no colour masking in HSV, no perceptual uniformity in LAB , and no chrominance suppression in YCbCr. I addressed this by going back and specifying each tab's educational purpose explicitly before even requesting the code.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?
When planning my part of the annotated code walkthrough, I consulted Claude on what content the notebook should include and why. The response was broad and covered more than was appropriate for my section, mixing in concepts that belonged to my teammates' colour spaces. I read through and filtered the output before realising I should have scoped the prompt to HSV specifically to begin with. I learnt that the context matters as much as the question, and giving an unscoped prompt returned an unscoped answer.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

Once the simulator's features were decided, the actual code implementation was fast. This meant my team could focus on what to build and why, rather than getting stuck on how. We were not slowed down by details like slider alignment or channel decomposition sizing. Our time went toward testing the simulator, verifying the conversion logic was correct, and identifying improvements. The division of labour between design thinking and implementation was what made the difference.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Why a minimum saturation threshold of approximately 30 is the accepted **starting value** when filtering out unstable hue regions in OpenCV. We can explain what the threshold does, but not why that specific value is the conventional starting point. The value of 30 is treated as a rule of thumb rather than something derived from the mathematics of the HSV model.

Kyra Talbot - 0391806L

Section 1: AI Tools Declared

ChatGPT – Used to explain YCbCr concepts, refine code walkthrough structure, debug Python/OpenCV issues, generate quiz questions, and support writing and reflection for the assignment.

Claude (Anthropic, Sonnet 4.6) – Used alongside ChatGPT to cross-check explanations, improve clarity of written sections, and help refine wording for technical descriptions and reflections.

NotebookLM – Used to summarise and understand academic papers related to YCbCr, and to extract key ideas for the study notes.

Google Scholar – Used to search for academic papers on YCbCr, RGB colour models, and skin segmentation techniques, and to locate credible research sources for citation.

Section 2: Notable Prompts

Prompt 1: *"If I were a lecturer, how could I possibly explain the topic of YCbCr to a class of students specifically focusing on these topics: luminance vs chrominance, converting from/to RGB, chroma subsampling, and is there anything important I should add?"*
After doing some research, I already had a basic understanding of what YCbCr is, but I wanted to ensure I included all important concepts and structured the explanation in a logical and understandable way. Through this prompt, I learnt about additional applications such as skin detection using YCbCr, which I then added to both the notes and code walkthrough. The response also helped introduce each section in a clearer way, which I later reworded in my own words to ensure I fully understood the concepts before presenting them.
Prompt 2: *"How can I make the code walkthrough as understandable as possible and how should I structure the code?"*
The response suggested organising the notebook into clear sections with short explanations followed by demonstrations. It also recommended using visual comparisons between RGB and YCbCr channels, chroma subsampling outputs, and skin detection examples to improve readability and educational value. This helped make the walkthrough feel more like a guided learning resource rather than just raw code output.
Prompt 3: *"I have a GitHub with my teammates but after I committed my code it's telling me 'Invalid Notebook - The Notebook Does Not Appear to Be Valid JSON'. How can I fix this issue?"*
The response explained that the Jupyter notebook became corrupted during a Git merge conflict because notebook files are stored using JSON formatting. The guidance focused on recovering earlier notebook versions from Git history, resolving merge conflicts safely, and avoiding manual notebook merges in the future. With the help of my teammates, I was then able to correctly restore and commit the notebook without losing any work.

Prompt 4: *<u>"Based on my YCbCr notes, I need to come up with questions to ask for a quiz with 4 multiple choices. For the correct answer, I need to justify why it's correct and explain why the other 3 are plausible pitfalls."</u>*
The response helped generate quiz questions on YCbCr fundamentals and explained why incorrect answers could still appear believable. This improved the educational quality of the quiz by encouraging understanding rather than simple memorisation. I eventually decided to use the YCbCr vs RGB question and the chroma subsampling question because I felt they covered the two most important concepts in the topic.

Section 3: 5 Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The hardest concept to explain was chroma subsampling, especially when trying to connect it to what the Cb and Cr channels actually represent. Understanding that Cb and Cr do not store actual "blue" or "red" images, but instead store colour differences relative to luminance, was already slightly unintuitive at first. On top of that, the notation used for chroma subsampling (such as 4:4:4, 4:2:2, and 4:2:0) was difficult to explain clearly, especially when describing what "half horizontal resolution" or "shared colour information across pixels" actually means in practice. It took visual examples and experimentation in code to properly understand how chrominance is reduced while luminance remains fully preserved.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?
One example was during the visualisation of the Cb and Cr channels in the code walkthrough. Initially, the output displayed the channels either in grayscale or using mixed red-blue colour maps. While these outputs were technically valid, I found them confusing and not very intuitive for students trying to understand the purpose of each chroma channel. To improve clarity, I modified the visualisation so that the Cb channel appeared mainly blue and the Cr channel mainly red. I also added an explanation clarifying that these were only visual representations rather than the actual stored colours.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

A specific moment where AI wasted time was during a GitHub merge conflict involving the Jupyter notebook. The notebook became corrupted, and GitHub reported that it was invalid JSON. At first, I followed several troubleshooting attempts with the merge codes, which still did not fully solve the issue. I eventually learned that manually merging notebook files is risky and that it is safer to restore a previous working version and reapply changes cleanly instead of trying to patch the JSON structure directly. Since the merge had already created several issues, I ultimately sent my completed code to a teammate who manually integrated it into the notebook to avoid further corruption.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

AI significantly improved the quality and speed of the YCbCr code walkthrough. It helped generate ideas for demonstrations such as chroma subsampling comparisons, skin detection using Cb and Cr thresholds, and incorrect conversion standards causing colour casts. These were concepts I wouldn't have known how to implement myself. The biggest difference was that AI allowed rapid iteration between explanation, coding, testing, and refinement, which helped transform the notebook from a basic technical demonstration into a more interactive learning resource.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

I think the part that my team doesn't feel confident answering is the reason why we add offsets in the YCbCr conversion, specifically the +128 applied to Cb and Cr and the +16 applied to Y. Although we understand that these offsets are used to fit the values into an 8-bit range, we still don't fully understand why those exact values were chosen. We can apply the conversion formulas correctly and understand what they do in practice, but explaining the deeper reasoning behind why chrominance is centred this way is still not completely clear to us.

Rebecca Hayward - 0148793A

Section 1: AI Tools Declared

NotebookLM (Google) — Used to upload and interrogate academic papers on colour spaces, extracting specific information about LAB fundamentals, conversion pipelines, and perceptual uniformity without reading every paper in full.

Claude (Anthropic, Sonnet 4.6) — Used throughout the project to understand academic concepts related to LAB, structure presentation slides, check factual accuracy of equations, answer conceptual questions about colour theory, and clean up syntax errors in code.

Claude Code (Anthropic) — Used specifically to clean up syntax errors and formatting issues in the Jupyter notebook code cells, catching problems that were not immediately obvious when writing manually.

Section 2: Notable Prompts

Prompt 1: *"What do these papers say about why LAB is considered perceptually uniform and what makes it different from RGB and HSV?"*

Used in NotebookLM after uploading the research papers. The response pulled specific explanations from the sources about opponent colour theory and how equal numerical distances in LAB space correspond to equal perceived differences, something neither RGB nor HSV can claim. This directly shaped the fundamentals section of the presentation and gave me a citable basis for the perceptual uniformity explanation rather than relying on my own interpretation.

Prompt 2: *"Do any of the papers explain what actually goes wrong visually if you skip gamma linearisation before converting RGB to LAB?"*

Used in NotebookLM to find source-backed evidence for the linearisation failure case. The response identified that the Muratbekova source specifically described the visual consequence -compressed shadows and clipped highlights ,which gave me concrete language for the failure cases slide rather than describing the error in purely mathematical terms.

Prompt 3: *"I understand the LAB conversion pipeline in theory but I am struggling to explain why XYZ is a necessary intermediate step and not just an extra complication. Can you help me explain this simply?"*

Asked to Claude when preparing speaker notes. Claude explained that XYZ exists as a device-independent anchor point, it represents human vision itself rather than any particular device, and that LAB cannot be derived directly from RGB because RGB is device-specific. This reframing made the pipeline explanation click and directly became the basis for how I introduced the conversion slide during the presentation.

Prompt 5: *"Can you check this slide for factual accuracy"* (pasted my RGB → LAB conversion slide content)

Claude identified that the slide showed only the XYZ → LAB equations but implied they converted directly from RGB, which would mislead the audience. It suggested adding a clarifying line noting that RGB must first pass through XYZ. A small fix but one I would not have caught myself because I already knew the full pipeline — the slide made sense to me but would not have to someone seeing it fresh.

Prompt 4: *"Can you go through this code cell and fix any syntax errors without changing the logic"* (pasted a LAB conversion notebook cell)

Used with Claude Code when a conversion cell had inconsistent indentation and a mismatched bracket that was not causing an obvious error but was making the code harder to read and review. Claude Code corrected the formatting cleanly without touching the mathematical logic. This was faster than manually scanning through line by line and meant I could focus on verifying the output was correct rather than whether the code was readable.

Section 3: 5 Questions

1. **Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?**

   The hardest concept to explain was ΔE*, both what it is and why it belongs in a discussion about LAB specifically. It was not immediately obvious to me why ΔE* is a LAB concept rather than a general colour comparison tool, and explaining why the formula only produces meaningful results in a perceptually uniform space took several attempts to phrase clearly. The RGB → LAB conversion pipeline was also difficult because the five steps feel arbitrary until you understand why each one exists, and compressing that into something a listener could follow in under a minute without losing the reasoning was genuinely challenging.

2. **Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?**

   When I asked Claude for the RGB → LAB conversion equations to put on a slide, it gave me the XYZ → LAB step only. The equations themselves were correct, but presenting them under the title "RGB → LAB Conversion" implied a direct single-step conversion that does not exist. I only caught it because I double-checked against the pipeline diagram. I added a clarifying note on the slide and learned to check not just whether the content is accurate but whether the framing around it is accurate too.

3. **Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?**

A frustrating pattern throughout the project was Claude giving contradictory advice on what to include in the code walkthrough. At different points it would suggest including a ΔE* demonstration, then in a later session either omit it entirely or suggest it was out of scope for a walkthrough and better suited to the written notes.Because I was working across multiple sessions without a consistent brief, I ended up going back and forth on the structure more than I should have. I eventually had to make a fixed decision on the scope myself and stop asking Claude to weigh in on it, which is what I should have done earlier. The lesson was that AI does not hold memory between sessions and will reason differently each time depending on how the question is framed, so using it to make structural decisions leads to inconsistency, it is better used to execute decisions you have already made yourself as well as consulting other sources to be absolutely sure.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

The most useful moments were during the code walkthrough, the presentation, and the short notes. At one point several people had pushed conflicting versions of the notebook without realising, and nobody was sure which version was correct or what had been overwritten. Claude helped us work through what had changed and why, which got the group back on track faster than it would have taken to sort manually. For the presentation, Claude helped with formatting ideas and structure when it was not clear how to lay out dense technical content visually. For the short notes it was particularly valuable early on when the whole task felt overwhelming — having Claude help break the content into sections made it feel manageable and gave us a starting point to work from and correct rather than starting from nothing.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Why the YCbCr luminance weights,0.299, 0.587, and 0.114have those specific values. The notes explain what they do and we implemented them correctly, but none of us could explain from first principles where those numbers actually come from or what the biological justification behind them is. We understood the standard well enough to use it, but not well enough to fully defend it.

Mia Busuttil - 0150406L

Section 1: AI Tools Declared

NotebookLM (Google): Used NotebookLM to upload the research papers I wanted to base my RGB documentation on and used the built in chat to help me write my documentation, generate diagrams and pictures for my powerpoint and generate questions for the quiz

Chat GBT: Used to structure study notes, helped create the introduction, conclusion and the RGB section specifically answering questions about key terms and the use of RGB.

Section 2: Notable Prompts

Prompt 1: "why does rgb exist"

Used to establish the fundamental motivation for using color in digital imaging. The resulting explanation highlighted how color is a powerful descriptor that simplifies object identification and takes advantage of the human ability to discern thousands of color shades compared to only two dozen shades of gray. This provided the necessary context to understand why RGB is the foundation of digital hardware, from scanners to monitors.

Prompt 2: "give me the why and how of rgb as well as an analogy"

Used to bridge the gap between biological reasoning and technical implementation. This prompt led to a detailed breakdown of the 24-bit "true color" system and the "Three Spotlights" analogy, which visualizes the additive color process. It clarified how three separate grayscale-like images are mathematically combined on a phosphor screen to produce a single composite color image.

Prompt 3: "what are the failure cases and limitations of rgb"

Used to identify technical trade-offs and weaknesses for a professional PowerPoint presentation. This query uncovered critical issues such as high channel correlation, which makes RGB "wasteful of bandwidth" for image compression, and perceptual non-uniformity, where mathematical shifts in color values do not align perfectly with human visual sensitivity. It also highlighted the "CRT-centric" nature of older standards like sRGB, which may not match modern flat-panel display characteristics.

Prompt 4: "generate and rgb graphic for the front page of my powerpoint"

Used as an explicit directive to create a visual artifact for a title slide. This prompt triggered the creation of a landscape-oriented infographic that summarized the 3D Cartesian colorcube and the additive mixing of red, green, and blue primary spectral components. By focusing on "true color" depth and hardware foundations, the resulting graphic served as a high-level visual summary of the entire learning pack

Prompt 5: “Can you give me 5 points that should be mentioned in my introduction and conclusion based on this paper”

This question generated important points that helped me create the introduction and conclusion

Section 3: 5 Questions

1. **Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?**

   The concept in my topic I find hardest to explain to someone else is how RGB works. RGB world on a Cartesian coordinate system and when trying to understand it I found it hard to visualize the coordinates and they are three dimensional, that is why I would find it hard to explain it to someone else. The spectrum the Cartesian coordinate system creates is over 16 million distinct colours. I find it hard to comprehend how the coordinates generate so many different colours, therefore I would find it hard to explain the concept.

2. **Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?**

   When asking NotebookLM to generate an infographic the first one it generated had an incorrect statement on it. This could have easily been missed considering I asked it to generate the diagram based on research papers and data already entered into the student notes To solve this problem I asked NotebookLM to regenerate the same diagram without the false statement.

3. **Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?**

   A specific moment where AI wasted time in this project was during the repeated failed attempts to generate complex technical infographics. According to the notebook state, two separate requests for highly detailed diagrams—one for the "Front Page Graphic" and another for "Failure Cases and Limitations"—resulted in a "Failed" status.

   Therefore the main problem I encountered with my use of AI was in generating info graphics and found no problems when it came to generating data and definitions as I used direct questions and entered relevant research papers into NotebookLM.

4. **Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made**
   The AI genuinely made the team faster by rapidly transforming raw academic data into the organized student notes format. The difference was the ability to submit research that was relevant to the project and have the AI instantly categorise it into a functional learning guide.

   This speed was evident in these specific areas:

- Creating Definitions: The AI took complex hardware descriptions from the sources and simplified them into the Sensor, Unrendered, and Rendered RGB categories.

- Validating Technical Data: It quickly pulled specific figures, such as the 16,777,216 distinct colours possible in a 24-bit true colour image, ensuring your "How RGB Works" section was accurate
- Creative Analogies: It integrated the "Three Spotlights" analogy from our chat to explain the additive process, making the "dimmer switch" concept (0 to 255 levels) intuitive for students.

5. **What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?**

   Even after completing this learning pack, our team still does not feel fully confident answering how modern AI systems automatically determine the “best” colour space for a specific computer vision task under changing real-world conditions. While we understand the strengths and weaknesses of RGB, HSV, YCbCr, and LAB individually, it remains challenging to predict which space will consistently perform best when factors such as lighting variation, compression artefacts, shadows, and hardware limitations interact at the same time in real-time AI systems.